

A disproof of Conjecture 00000002416

No open subgroup of $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ is finitely generated

1 The conjecture

Conjecture 00000002416 reads (English text; the Chinese text says the same):

Conjecture: The spectrum of the minimal number of generators of open subgroups of $G_{\mathbb{Q}}$ is $\{2, 3\}$; the density of $d = 3$ subgroups is 0 (an explicit density – almost all subgroups are 2-generated); the Shafarevich spectrum is complete.

The statement is a conjunction of three claims separated by semicolons. We refute the first one, which the second one also presupposes (it speaks of the “ $d = 3$ subgroups”). Hence the conjecture is false.

2 Reading and definitions

- $G_{\mathbb{Q}} = \text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ is the absolute Galois group of $\mathbb{Q}$, with its Krull topology. In Lean this is Mathlib’s `Field.absoluteGaloisGroup Q` (with Lean’s $\mathbb{Q}$ written as `Q` here), the group of $\mathbb{Q}$ -algebra automorphisms of `AlgebraicClosure Q` with the Krull topology.
- A subset S of a topological group H *topologically generates* H if the subgroup $\langle S \rangle$ generated by S is dense in H (Lean: `TopGenerates`).
- The *minimal number of generators* of H is $d(H) = \inf\{|S| : S \subseteq H \text{ finite, } S \text{ topologically generates } H\} \in \mathbb{N} \cup \{\infty\}$, with $d(H) = \infty$ (Lean: `⊤` in `ENat`) when no finite set topologically generates H (Lean: `minGen`).
- An open subgroup $U \leq G_{\mathbb{Q}}$ carries the subspace topology. The *spectrum* is $\{d(U) : U \leq G_{\mathbb{Q}} \text{ an open subgroup}\} \subseteq \mathbb{N} \cup \{\infty\}$ (Lean: `genSpectrum`). It contains $d(G_{\mathbb{Q}})$, since $G_{\mathbb{Q}}$ is an open subgroup of itself.
- We also treat abstract generation: $d_{\text{abs}}(H)$ is the least size of a finite S with $\langle S \rangle = H$ (Lean: `minGenAbs`, spectrum `genSpectrumAbs`).

3 The result

Theorem 1. *No open subgroup of $G_{\mathbb{Q}}$ is topologically finitely generated. Hence $d(U) = \infty$ for every open subgroup U , the spectrum is $\{\infty\}$, and in particular it is not $\{2, 3\}$; neither 2 nor 3 belongs to it. The same holds for abstract generation.*

Step 1: quadratic characters

For a prime p fix $\sqrt{p} \in \overline{\mathbb{Q}}$ with $(\sqrt{p})^2 = p$ (any choice). For $\sigma \in G_{\mathbb{Q}}$ we have $\sigma(\sqrt{p})^2 = \sigma(p) = p$, so $\sigma(\sqrt{p}) = \pm\sqrt{p}$. Put $\chi_p(\sigma) = 1$ if $\sigma(\sqrt{p}) = \sqrt{p}$ and $\chi_p(\sigma) = -1$ otherwise. Because $\sqrt{p} \neq -\sqrt{p}$ (characteristic 0, $p \neq 0$), a case check shows that $\chi_p : G_{\mathbb{Q}} \rightarrow \{\pm 1\} = \mathbb{Z}^\times$ is a group homomorphism.

Its kernel contains $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{p}))$, the fixing subgroup of the finite extension $\mathbb{Q}(\sqrt{p})$, which is open in the Krull topology. A subgroup containing an open subgroup is open, so $\ker \chi_p$ is open.

Step 2: distinct primes give distinct characters

Let $p \neq q$ be primes with $\chi_p = \chi_q$. For every σ , $\sigma(\sqrt{p}\sqrt{q}) = \chi_p(\sigma)\chi_q(\sigma)\sqrt{p}\sqrt{q} = \sqrt{p}\sqrt{q}$. The extension $\overline{\mathbb{Q}}/\mathbb{Q}$ is Galois, so by the Galois correspondence for (possibly infinite) Galois extensions the elements fixed by all of $G_{\mathbb{Q}}$ form $\mathbb{Q}$. Hence $\sqrt{p}\sqrt{q} = r \in \mathbb{Q}$ and $r^2 = pq$. A natural number that is a square in $\mathbb{Q}$ is a square in $\mathbb{N}$, so $pq = m^2$. Then $p \mid m$, say $m = pk$, so $q = pk^2$ and $p \mid q$, forcing $p = q$, a contradiction.

Step 3: counting

Let $f, g : H \rightarrow M$ be homomorphisms of groups, H a topological group, with open kernels, and suppose $f = g$ on a set S that topologically generates H . The equalizer $E = \{x : f(x) = g(x)\}$ is a subgroup containing the open subgroup $\ker f \cap \ker g$, so E is open, hence closed. It contains $\langle S \rangle$, so it contains its closure, which is H . Thus $f = g$.

Suppose a finite set T topologically generates $G_{\mathbb{Q}}$. The map sending a prime p to the restriction $\chi_p|_T \in \{\pm 1\}^T$ is injective by Step 3 and Step 2. The primes are infinite and $\{\pm 1\}^T$ is finite, a contradiction. So $G_{\mathbb{Q}}$ is not topologically finitely generated.

Step 4: open subgroups

$G_{\mathbb{Q}}$ is compact. Let U be an open subgroup and suppose a finite $S \subseteq U$ topologically generates U in the subspace topology. The cosets gU form an open cover of $G_{\mathbb{Q}}$, so finitely many, $r_1U, \dots, r_kU$, cover it. Let $H = \langle S \cup \{r_1, \dots, r_k\} \rangle \leq G_{\mathbb{Q}}$. Since the inclusion $U \rightarrow G_{\mathbb{Q}}$ is continuous, U lies in the closure $\overline{H}$; the r_i lie in H ; and $\overline{H}$ is a subgroup, so it contains every r_iu and hence equals $G_{\mathbb{Q}}$. So the finite set $S \cup \{r_i\}$ topologically generates $G_{\mathbb{Q}}$, contradicting Step 3.

Finally, $\langle S \rangle = U$ implies that $\langle S \rangle$ is dense in U , so the abstract version follows. Since $G_{\mathbb{Q}}$ itself is open, the spectrum is exactly $\{\infty\}$. $\square$

4 Lean formalization

The Lean file `Conjecture2416/Basic.lean` (296 lines, only import `Mathlib`, namespace `C2416`) contains:

- `topFG_of_open`: Step 4 for any compact topological group.
- `eq_of_eqOn_gen`: Step 3, the equalizer argument.
- `sqrtQ`, `chi`, `chiHom`, `chiHom_ker_isOpen`: Step 1.
- `mem_range_of_fixed` (from `Mathlib's InfiniteGalois.fixedField_fixingSubgroup`), `not_isSquare_mul`, `chiHom_injective`: Step 2.

- `gal_not_topFG`, `absGal_not_topFG`, `open_subgroup_not_topFG`, `open_subgroup_not_fg`, `minGen_open_subgroup`, `genSpectrum_absGal`, `genSpectrumAbs_absGal`.

The main theorem is

```
theorem conjecture_2416_false :
  genSpectrum (Field.absoluteGaloisGroup Q) != {2, 3} /\
  2 not-in genSpectrum (Field.absoluteGaloisGroup Q) /\
  3 not-in genSpectrum (Field.absoluteGaloisGroup Q)
```

(with the Lean symbols for not-equal, and, not-in written in ASCII), together with `conjecture_2416_false_abstract`, the same statement for `genSpectrumAbs`, and `genSpectrum_absGal` : `genSpectrum (Field.absoluteGaloisGroup Q) = {top}`.

One technical point: `AlgebraicClosure Q` carries two $\mathbb{Q}$ -algebra structures that are equal but not reducibly so (`AlgebraicClosure.instAlgebra`, used in the definition of `Field.absoluteGaloisGroup`, and `DivisionRing.toRatAlgebra`). The file gives the former local priority, so all statements refer to the same automorphism group as `Field.absoluteGaloisGroup Q`.

Axiom audit. `lake build` succeeds. `#print axioms` for `conjecture_2416_false`, `conjecture_2416_false_abstract`, `genSpectrum_absGal` and `open_subgroup_not_topFG` reports only `propext`, `Classical.choice`, `Quot.sound`. There is no `sorry` and no `native_decide`.

5 Scope

We refute the first clause for $G_{\mathbb{Q}} = \text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$, with topological generation (the standard notion for profinite groups) and with abstract generation. We do not address the other two clauses, which are not needed: the conjecture is their conjunction with the first one.